AI-Based Email Assistant

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1. Abstract

This project automates email replies using Artificial Intelligence and Natural Language Processing techniques. It reads user emails, detects intent, and generates smart, professional replies automatically. The assistant is trained to improve over time, adapt to different communication tones, and handle diverse scenarios efficiently.

2. Introduction

Emails are essential in professional communication, but manual responses are time-consuming. This project aims to simplify that by introducing automation with AI, ensuring faster and better-quality communication while maintaining user control.

3. Literature Survey / Review

Previous works like ‘Intelligent Email’ (Dredze et al., 2008) and Miura et al. (2025) laid the foundation for automated replies, but lacked contextual depth and adaptability. Our system bridges that gap using transformer models and reinforcement feedback.

4. Methodology / Proposed Work

The AI assistant follows five steps: preprocessing, semantic understanding, clarification question generation, drafting, and final response creation. NLP models like BERT and GPT

ensure contextual relevance and linguistic precision.

5. System Design / Architecture

The modular system uses microservices with separate layers for preprocessing, classification, summarization, and drafting. The backend is built on Spring Boot with real-time integration through IMAP/SMTP. The architecture ensures scalability and reliability.

6. Expected Outcome / Results

Expected results include 70% faster email handling, higher accuracy in reply tone, and reduced cognitive load. The system improves with user feedback, ensuring personalization over time.

7. Applications

Usable in corporations, education, and customer service, the assistant handles FAQs, scheduling, and updates automatically. It ensures productivity while maintaining communication standards.

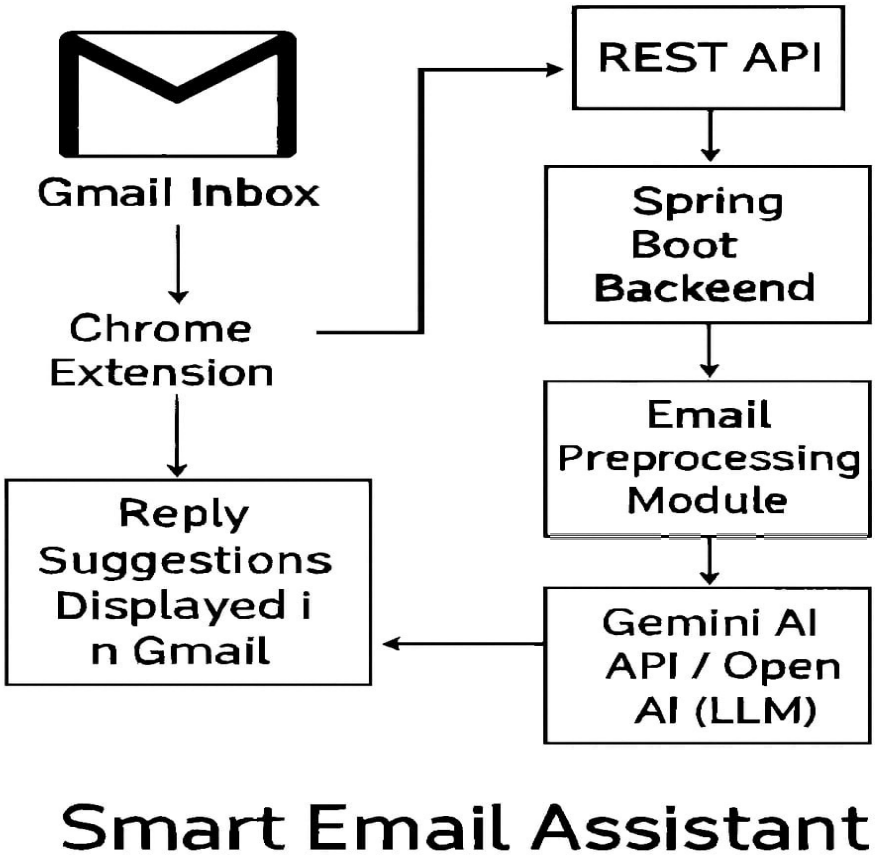

Figure: System Architecture - Smart Email Assistant

8. Scope of the Project

Currently focused on formal communication, with future expansion into multilingual support, sentiment-aware replies, and integration with virtual meeting tools.

9. Future Enhancements

Upcoming updates include predictive message prioritization, emotion recognition, and voice-interactive responses. Cloud scalability and security enhancements are planned.

10. Conclusion

The Smart Email Assistant represents a step forward in automating professional communication. It integrates modern AI technologies to enhance workflow, reduce effort,

and improve accuracy.

11. References

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